

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Digital Learning Handout

Part A: Content Design

Course Title	Preprocessing for AI-Ready Data
Course No(s)	DSE ZG505
Credit Units	4
Credit Model	3-1-0
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Course Description:

AI-readiness of data across modalities, e.g., audio, text, images, graphs, spatial, time-series; data wrangling (munging); contextual enrichment of metadata; feature selection; modality-specific feature extraction techniques for different data modalities; multimodal feature extraction; knowledge graph; pre-trained deep learning model-based feature extraction; pre-processing for deep learning; for data governance, e.g., fairness, explainability, security and privacy; regulatory compliance.

Course Objectives

No	Course Objective
CO1	Diagnose data quality issues (missing values, duplicates, outliers, inconsistent schemas, label noise) and select appropriate remediation strategies
CO2	Apply feature engineering, encoding, and scaling techniques suited to different model families
CO3	Pre-process unstructured and multimodal data (text, images, audio, graphs, spatial, time series) into model-ready representations
CO4	Handle imbalanced datasets and evaluate the trade-offs of resampling and reweighting strategies
CO5	Design pre-trained deep learning model-based feature extraction
CO6	Evaluate data pre-processing decisions through the lens of data governance: fairness, explainability, security, privacy, and regulatory compliance

Text Book(s):

T1	Feature Engineering for Machine Learning, O'Reilly, 2018, Alice Zheng & Amanda Casari
T2	Data Cleaning, ACM Books, 2019, Ihab F. Ilyas & Xu Chu
T3	Deep Learning (Adaptive Computation and Machine Learning series), MIT Press, 2016, Ian Goodfellow, Yoshua Bengio & Aaron Courville
T4	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly, Aurélien Géron

Reference Book(s) & other resources:

R1	Data Science, MIT Press, John D. Kelleher & Brendan Tierney
R2	Statistical Analysis with Missing Data, Wiley, Roderick J.A. Little & Donald B. Rubin
R3	Flexible Imputation of Missing Data, Stef van Buuren
R4	Outlier Analysis, Springer, Charu C. Aggarwal
R5	Data Matching, Springer, Peter Christen
R6	Feature Engineering and Selection, CRC Press, Max Kuhn & Kjell Johnson
R7	Speech and Language Processing, Daniel Jurafsky & James H. Martin
R8	Computer Vision: Algorithms and Applications, Richard Szeliski
R9	Forecasting: Principles and Practice, Rob J. Hyndman & George Athanasopoulos (free online)
R10	Fairness and Machine Learning, Solon Barocas, Moritz Hardt & Arvind Narayanan (fairmlbook.org)
R11	Research Papers, Blogs, and Tool Documentation (Great Expectations, scikit-learn, spaCy, PyTorch, TensorFlow docs; SMOTE, LOF, Isolation Forest, Box-Cox, Datasheets for Datasets, Model Cards, BPE, GAN, DDPM and related papers)
R12	Python Data Science Handbook, O'Reilly, Jake VanderPlas
R13	Time Series Analysis: Forecasting and Control, Wiley, George E.P. Box, Gwilym M. Jenkins, Gregory C. Reinsel & Greta M. Ljung
R14	Graph Representation Learning by William L. Hamilton

Learning Outcomes: Students will be able to

LO1	Assess and remediate data quality issues across structured, unstructured, and multimodal datasets
LO2	Engineer, encode, and scale features appropriately for different model families, including modality-specific and multimodal feature extraction
LO3	Build model-ready representations for text, image, audio, graph, spatial, and time-series data, including deep-learning-specific pipelines
LO4	Apply LLM-based and pre-trained deep learning model-based techniques for feature representation and data augmentation
LO5	Evaluate pre-processing pipelines for data governance: fairness, explainability, security, privacy, and regulatory compliance

Modular Content Structure

1. Foundations of AI-Ready Data & the Data Lifecycle

- What makes data "AI-ready"?
- Quality dimensions: accuracy, completeness, consistency, timeliness, validity, uniqueness
- The data pipeline
- Structured vs. unstructured vs. semi-structured data vs. multimodal data (types/examples of multimodal data).
- Relationship to Introduction to Data Science (concurrent course)

2. Data Quality Assessment & Profiling

- Data profiling techniques; summary statistics, distributions, cardinality checks
- Defining data quality metrics and thresholds

3. Handling Missing Data & Outlier Detection

- Deletion strategies vs. imputation
- Missingness as a feature; flag-and-fill approach
- Statistical methods (Z-score, IQR), model-based methods (Isolation Forest, LOF, DBSCAN)
- Contextual vs. global outliers

4. Data Cleaning: Duplicates, Inconsistencies & Entity Resolution

- Exact vs. fuzzy duplicate detection
- Standardizing formats
- Record linkage and entity resolution basics

5. Feature Encoding

- One-hot, label, ordinal, target/mean, frequency, and binary encoding
- Hashing for high-cardinality categorical variables
- Encoding pitfalls

6. Feature Scaling, Normalization & Transformation

- Min-max scaling, standardization, robust scaling
- Log, Box-Cox, and power transforms for skewed distributions
- Pros and Cons of scaling matters

7. Feature Engineering & Selection

- Domain-driven feature creation
- Dimensionality reduction, feature importance, mutual information
- Filter, wrapper, and embedded feature selection methods
- Dimensionality reduction in the AI-readiness context

8. Handling Imbalanced Data

- Oversampling (SMOTE, ADASYN), undersampling, hybrid methods
- Cost-sensitive learning and class weighting
- Evaluation pitfalls: accuracy trap; appropriate metrics

9. Text Pre-processing & NLP-Ready Pipelines

- Cleaning: camelcasing, punctuation, spelling, emojis
- Tokenization, normalization, stop word removal, stemming, lemmatization
- Vectorization: Bag-of-Words, TF-IDF, embeddings
- Handling noisy text

10. Image & Signal Pre-processing for AI

- Resizing, normalization, augmentation
- Denoising, contrast enhancement, handling class imbalance in vision datasets
- Basics of audio/signal pre-processing

11. Time-Series Pre-processing

- Resampling, interpolation, detrending, seasonality decomposition
- Handling irregular timestamps and lag feature creation
- Windowing strategies and avoiding temporal data leakage

12. Graph Data Pre-processing & Knowledge Graphs

- Graph data representation
- Data cleaning for graph datasets
- Constructing graphs from raw/relational data
- Knowledge graph construction
- Graph feature extraction: node2vec/DeepWalk and graph neural network embeddings

13. Multimodal Feature Extraction

- Feature extraction across modalities
- Fusion strategies: early, late, and hybrid fusion
- Cross-modal embeddings
- Challenges in multimodal learning

14. Pre-processing for Deep Learning

- How pre-processing needs differ for deep learning vs. classical ML: less manual feature engineering, more emphasis on scale, normalization, and pipeline throughput
- Input normalization/standardization conventions for neural nets
- Data augmentation as a deep learning-native pre-processing step
- Sub-word tokenization for deep NLP

15. Generative AI-Based Techniques for Data Pre-processing

- Synthetic data generation and diffusion models for augmentation and class balancing
- LLM-based data augmentation: paraphrasing, synthetic labeling, programmatic data cleaning and normalization
- Generative imputation and anomaly correction as an alternative to classical imputation
- Evaluating synthetic data quality: distributional fidelity, mode collapse, privacy leakage, downstream utility
- Course synthesis: revisiting the pre-processing checklist across classical, deep learning, and generative techniques

16. Bias, Fairness & Ethical Considerations

- How pre-processing choices introduce or amplify bias
- Privacy-preserving pre-processing
- Documentation practices
- Regulatory compliance considerations

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Foundations of AI-Ready Data & the Data Lifecycle	What makes data "AI-ready"? Quality dimensions: accuracy, completeness, consistency, timeliness, validity, uniqueness; The data pipeline: ingestion → cleaning → transformation → feature engineering → validation → serving; Structured vs. unstructured vs. semi-structured data vs. multimodal data (types/examples of multimodal data). Relationship to Introduction to Data Science (concurrent course)	R1 Ch.1–2; T4 Ch.2; T2 Ch.1
2	Data Quality Assessment & Profiling	Data profiling techniques; summary statistics, distributions, cardinality checks; Automated profiling tools (pandas-profiling/ydata-profiling, Great Expectations); Defining data quality metrics and thresholds	T2 Ch.2; R11 (Great Expectations & profiling docs)
3	Handling Missing Data & Outlier Detection	Deletion strategies vs. imputation (mean/median/mode, KNN, regression); Missingness as a feature; flag-and-fill approach; Statistical methods (Z-score, IQR), model-based methods (Isolation Forest, LOF, DBSCAN); Contextual vs. global outliers; when to remove vs. cap vs. transform	R2 Ch.1–3; R3 Ch.1–2; R12 R4 Ch.1–2; R11 (LOF, Isolation Forest papers)
4	Data Cleaning: Duplicates, Inconsistencies & Entity Resolution	Exact vs. fuzzy duplicate detection (Levenshtein, Jaccard, embedding similarity); Standardizing formats (dates, units, categorical labels); Record linkage and entity resolution basics	T2 Ch.4–5; R5 Ch.1,5
5	Feature Encoding (Categorical Data)	One-hot, label, ordinal, target/mean, frequency, and binary encoding; Handling high-cardinality categorical variables; hashing trick; Encoding pitfalls: data leakage, unseen categories at inference	T1 Ch.5–6; R11 (scikit-learn preprocessing docs)
6	Feature Scaling, Normalization & Transformation	Min-max scaling, standardization, robust scaling; Log, Box-Cox, and power transforms for skewed distributions; When scaling matters (distance-/gradient-based models) vs. when it doesn't (tree-based models)	T1 Ch.2; R11 (Box & Cox, 1964)
7	Feature Engineering & Selection	Domain-driven feature creation, interaction terms, polynomial features; Dimensionality reduction: PCA, feature importance, mutual information; Filter, wrapper, and embedded feature selection methods; Dimensionality reduction in the AI-readiness context	T1 Ch.6; R6 Ch.3–5; R11 (Guyon & Elisseeff, 2003)
8	Handling Imbalanced Data	Oversampling (SMOTE, ADASYN), undersampling, hybrid methods; Cost-sensitive learning and class	R11 (SMOTE 2002; He & Garcia 2009;

Contact Session	List of Topic Title	Sub-Topics	Reference
		weighting; Evaluation pitfalls: accuracy trap; appropriate metrics (F1, PR-AUC)	imbalanced-learn docs)
9	Text Preprocessing & NLP-Ready Pipelines	Cleaning: camelcasing, punctuation, spelling, emojis Tokenization, normalization, stopword removal, stemming/lemmatization; Vectorization: Bag-of-Words, TF-IDF, embeddings (word2vec, transformer-based); Handling noisy text (social media, OCR output, multilingual text)	R7 Ch.2,6; T1 Ch.3–4; R11 (spaCy docs)
10	Image & Signal Preprocessing for AI	Resizing, normalization, augmentation (flip, rotate, crop, color jitter); Denoising, contrast enhancement, handling class imbalance in vision datasets; Basics of audio/signal preprocessing (sampling, filtering, spectrograms)	R8 Ch.3; T1 Ch.8; R11 (Shorten & Khoshgoftaar, 2019)
11	Time-Series Preprocessing	Resampling, interpolation, detrending, seasonality decomposition; Handling irregular timestamps and lag feature creation; Windowing strategies and avoiding temporal data leakage	R9 Ch.2–3; R13 Ch.1–2
12	Graph Data Pre-processing & Knowledge Graphs	Graph data representation: nodes, edges, attributes; Data cleaning for graph datasets; constructing graphs from raw/relational data; Knowledge graph construction: entity/relation extraction, linking, ontology alignment; Graph feature extraction: node2vec/DeepWalk and graph neural network embeddings	R14 Ch. 1-5
13	Multimodal Feature Extraction	Feature extraction across modalities; Fusion strategies: early, late, and hybrid fusion; Cross-modal embeddings; Challenges in multimodal learning	
14	Preprocessing for Deep Learning	How preprocessing needs differ for deep learning vs. classical ML: less manual feature engineering, more emphasis on scale, normalization, and pipeline throughput; Input normalization/standardization conventions for neural nets; Data augmentation as a deep learning-native pre-processing step; Sub-word tokenization for deep NLP	T3 Ch.11; R11 (Sennrich et al. 2016; PyTorch/TF docs)
15	Generative AI-Based Techniques for Data Preprocessing	Synthetic data generation and diffusion models for augmentation and class balancing; LLM-based data augmentation: paraphrasing, synthetic labeling, programmatic data cleaning and normalization; Generative imputation and anomaly correction as an alternative to classical imputation; Evaluating synthetic data quality: distributional fidelity, mode collapse, privacy leakage, downstream utility; Course synthesis: revisiting the preprocessing checklist across	R11 (Goodfellow et al. 2014; Ho et al. 2020; Jordon et al. 2022)

Contact Session	List of Topic Title	Sub-Topics	Reference
		classical, deep learning, and generative techniques	
16	Bias, Fairness & Ethical Considerations in Preprocessing	How preprocessing choices introduce or amplify bias (sampling bias, proxy variables, label bias); Privacy-preserving preprocessing: anonymization, pseudonymization, differential privacy basics; Documentation practices: datasheets for datasets, data provenance, consent tracking; Regulatory compliance considerations in data preprocessing	R10 Ch.1–2; R11 (Gebru et al. 2021; Mitchell et al. 2019; Suresh & Gutttag 2021)

Reference codes (T = Text Book, R = Reference Book/resource) correspond to the Text Book(s) and Reference Book(s) & other resources tables in Part A above. Chapter numbers follow standard/current editions; confirm against your library's specific printing.

Experiential Learning Components:

- Lab work: 15 (guided technical labs across all modules)
- Project work: 1 (Capstone Project)
- Case Study: Module 14 — discussion and real-world case studies on preprocessing failures
- Simulation: 0
- Work Integrated Learning Assignment: Midterm Practical Exam (Modules 1–8)
- Design work/Field work: 0

Objective of Experiential Learning Component:

Hands-on sessions applying data preprocessing techniques using industry-standard Python tooling, culminating in a reproducible, documented, end-to-end preprocessing pipeline for a real-world capstone dataset.

Scope of Experiential Learning Component:

Programming language: Python

Tools and libraries: pandas, NumPy, scikit-learn, Jupyter/Colab, SQL, spaCy/NLTK, OpenCV/Pillow, Apache Spark (intro), Great Expectations/pandas-profiling, PyTorch, TensorFlow

Lab Infrastructure:

Online / Open source / Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Explore a raw, messy dataset with pandas	1
2	Generate a data quality report and quality scorecard	2
3	Compare imputation strategies and their downstream effect on model performance	3
4	Detect and treat outliers in a financial/sensor dataset	4
5	Deduplicate and standardize a multi-source customer dataset	5

Lab No	Lab Objective	Session Reference
6	Build an encoding pipeline with scikit-learn ColumnTransformer	6
7	Compare model sensitivity to scaling choices	7
8	Engineer and select features for a tabular prediction task	8
9	Address class imbalance in a fraud/churn dataset	9
10	Build a text preprocessing pipeline with spaCy/NLTK for a downstream classifier	10
11	Build an image preprocessing/augmentation pipeline with OpenCV/Pillow	11
12	Prepare a multivariate time-series dataset for forecasting	12
13	Build a PyTorch/TensorFlow data pipeline with on-the-fly augmentation and batching	13
14	Generate synthetic data using Generative AI models and compare it with conventional data augmentation techniques across different data modalities	15

Evaluation Scheme:

Legend: EC = Evaluation Component

Evaluation Component	Name	Type	Weight	Duration	Day, Date, Session, Time
EC-1	Quiz (2), Assignments (2)	Online	30%	Ongoing	To be announced
EC-2	Midterm Exam (Modules 1–8 coverage)	Closed Book	30%	1.5 hours	19/09/2026 (AN)
EC-3	Comprehensive Exam	Open Book	40%	2 hours	05/12/2026 (AN)

Syllabus for Midterm Practical Exam (Closed Book): Modules 1 to 8

Syllabus for Comprehensive coverage (Open Book): All modules

Evaluation Guidelines:

- EC-1 consists of individually graded lab assignments (8 selected, from across the 15 guided labs). Deadlines and submission instructions will be posted on the course portal in a timely manner.
- For Closed Book tests: no books or reference material of any kind will be permitted.
- For Open Book components: "open book" means the prescribed text/reference books (publisher copy only) and does not include any other learning material.
- Students unable to attempt a Regular Test/Exam due to genuine exigencies should follow the institution's Make-Up Test/Exam procedure, as announced on the course portal.

It shall be the responsibility of the individual student to be regular in following the self-study schedule given in this course handout, attend the sessions, and complete all prescribed evaluation components — Lab Assignments, the Midterm Practical Exam, and the Capstone Project — according to the evaluation scheme provided above.